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Functional Near-Infrared Spectroscopy and Machine Learning For Neurocognitive and Mental-Health Screening

Literature Review

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1. Literature Review

This literature review focuses on functional near-infrared spectroscopy (fNIRS) and machine learning as techniques for researching neurocognitive and health-related task patterns. The updated methodology is intentionally modest, as fNIRS is taken here as a tool to measure relative haemodynamic changes, whereas machine-learning models depend on valid preprocessing, physiologically relevant signal representation, and leak-proof evaluation.

The cited evidence shows that modelling of task states, workload, walking-task, and cognitive-effort tasks has stronger grounds than classification for clinical diagnostics purposes. This means that the thesis does not imply that the fNIRS technique alone can detect mental disorders or neurodevelopmental conditions. Rather, it questions whether relative frontal haemodynamic patterns can be preprocessed and modelled in a valid manner.

1.1. Scope of This Literature Analysis

The key challenge is not only in obtaining high accuracy with a classifier. Is the signal entered into a classifier plausible? As noted in current fNIRS methodology, there are a number of physiological considerations which need to be accounted for to obtain a valid result [4, 5, 6].

Several papers cited in this review contain examples of fNIRS classification and deep learning [1, 2, 7, 3, 8, 9]. However, their relevance is conditional on the limitations of fNIRS data. Most evidence that the current thesis relies on concerns the issues of short-channel regression, physiological confound, benchmarking, and data leakage [10, 11, 12, 13, 14, 15].

2. Methodological Evidence From fNIRS Studies

2.1. GASF-Based Task-State Classification

An example of conversion of the fNIRS signals to GASF images is provided by Wickramaratne and Mahmud [1]. It is useful as a representative approach to signal representation in classification tasks, but should not be considered as the preferred methodology in this thesis due to potential issues. First of all, fNIRS provides a multichannel time series. Its conversion to an image could significantly alter its temporal structure.

It means that in this thesis, the use of images created from time-series based on GASF is considered as an example from the literature, and not as a preferred choice of signal representation. Whatever type of signal representation will be chosen for processing in this thesis, the window duration, sampling frequency, and channel distribution should be preserved. Otherwise, the representation becomes too generic to be valid for physiological data.

Gramian Angular Summation Field representation:

$$\tilde{x}_i = \frac{x_i - \min(X)}{\max(X) - \min(X)} \tag{1}$$

$$\varphi_i = \arccos(\tilde{x}_i) \tag{2}$$

$$\mathrm{GASF}_{ij} = \cos(\varphi_i + \varphi_j) \tag{3}$$

$$\mathrm{GASF}_{ij} = \tilde{x}_i \tilde{x}_j - \sqrt{1 - \tilde{x}_i^2}\sqrt{1 - \tilde{x}_j^2} \tag{4}$$

Note: Here are the formulas to convert an fNIRS time-series signal into GASF representation.

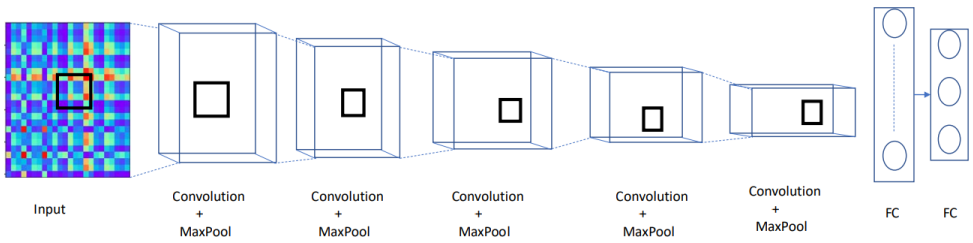


Fig. 1. Example convolutional neural network used for GASF-based fNIRS task classification [1].

2.2. Signal Representation With Preserved Temporal Structure

In this thesis, the primary input representation is a multichannel time series of haemodynamics. For each fixed-duration analysis window, the representation is a channel-by-time-by-signal three-

dimensional matrix:

$$X \in \mathrm{R}^{C \times T \times K}, \tag{5}$$

where C is the number of fNIRS channels, T is the number of time samples in the window, and K is the number of haemodynamic signals in each channel, for example ΔHbO, ΔHbR, HbDiff, or HbT. Time windows are predefined in physical time before training, and sampling structure is retained.

Modern deep-learning models for classification of fNIRS data show that CNN-LSTM, one-dimensional convolutional, attentional, and temporal convolutional models can process temporal and spatial aspects of fNIRS signal without changing its representation into an image [2, 9, 7]. Additionally, classical machine-learning models can use the extracted features from the input windows under a condition that those will be fitted and evaluated without leakage [14].

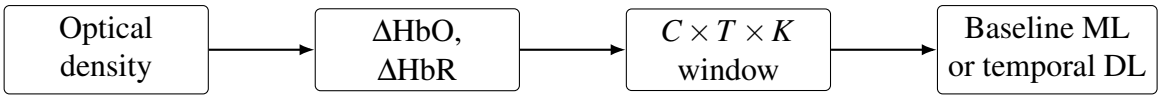


Fig. 2. Sequence-preserving fNIRS representation used for model input.

2.3. CNN + LSTM Neural Network Model for fNIRS Workload Classification

Convolutional and LSTM layers are necessary to create an architecture capable of processing a two-way variation in fNIRS data. The former learns local patterns, whereas the latter deals with dependencies between samples [2]. Nevertheless, the reported high accuracy of the fNIRS workload classifier cannot imply its diagnostic ability as the workload itself is determined experimentally and may reflect various factors.

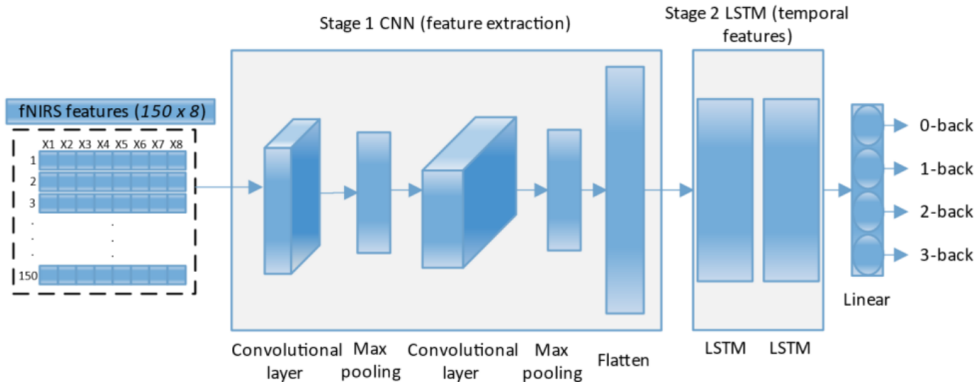


Fig. 3. Example CNN-LSTM architecture for fNIRS workload classification [2].

Convolution operation and related functions:

$$Y[i] = f\left(\sum_{j=1}^{k} X[i+j-1] \cdot W[j] + b\right) \tag{6}$$

$$Y[i] = \max(0, X[i]) \quad \text{(ReLU activation)} \tag{7}$$

$$Y[i] = \max(X[i \cdot p : (i+1) \cdot p]) \quad \text{(Max-pooling operation)} \tag{8}$$

Note: Deep architectures are considered only after reliable leakage-safe baselines are established.

2.4. Correct Haemodynamic Quantities

In continuous-wave fNIRS, haemoglobin concentration changes can be estimated through conversion of optical density to relative oxyhaemoglobin and deoxyhaemoglobin changes after applying the modified Beer-Lambert law [4, 16]. These haemodynamic quantities are measured relatively, not absolutely.

$$\Delta A(\lambda) = \mathrm{DPF} \cdot r \cdot [\varepsilon_{\mathrm{HbO}}(\lambda)\Delta c_{\mathrm{HbO}} + \varepsilon_{\mathrm{HbR}}(\lambda)\Delta c_{\mathrm{HbR}}] \tag{9}$$

$$\mathrm{HbDiff} = \Delta\mathrm{HbO} - \Delta\mathrm{HbR} \tag{10}$$

$$\mathrm{HbT} = \Delta\mathrm{HbO} + \Delta\mathrm{HbR} \tag{11}$$

The difference between relative oxyhaemoglobin and deoxyhaemoglobin changes is called HbDiff. Relative haemoglobin total changes is HbT. Both terms cannot be considered oxygenation indexes or oxygen saturation measures. Ratio-based values like the regional oxygen saturation or tissue saturation index are calculated explicitly by the device and software algorithms [16].

$$\mathrm{rSO_2 \text{ or } TSI} = \frac{\mathrm{HbO_2}}{\mathrm{HbO_2} + \mathrm{Hb}} \times 100 \quad \text{only when saturation is explicitly estimated.} \tag{12}$$

Thus, the present thesis defines ΔHbO, ΔHbR, HbDiff, and HbT as relative haemodynamic measures. Physiological interpretation is possible only in terms of task-related patterns under the mentioned preprocessing.

2.5. Physiological Confounds and Short-Channel Regression

fNIRS data from channels separated by large distances contain both haemodynamic signal originating in the cortex and haemodynamic activity caused by other factors [5, 12]. Those can be scalp blood flow, heart rate, respiration, blood pressure changes, Mayer waves, motion artifacts, etc. If those confounds are not removed, the classifier learns a superficial signal instead of actual task-related patterns.

One approach to addressing this problem is to use short-distance channels to estimate the effect of extracerebral haemodynamic activity. It is applicable since the present thesis database contains data for short-distance channels. Thus, those should be used as nuisance regressors in preprocessing.

$$y_{\text{clean}} = y_{\text{long}} - \alpha y_{\text{short}}, \qquad \alpha = \frac{\langle y_{\text{short}}, y_{\text{long}} \rangle}{\langle y_{\text{short}}, y_{\text{short}} \rangle}. \tag{13}$$

Another approach would be using the general GLM method where short- and long-distance channel regressors, and physiological confound regressors will be used. The cleaned haemodynamic signal will be obtained as residuals.

$$Y = X_{\text{task}}\beta_{\text{task}} + X_{\text{SDC}}\beta_{\text{SDC}} + X_{\text{phys}}\beta_{\text{phys}} + \varepsilon. \tag{14}$$

Several recent papers have demonstrated the benefits of short-channel correction [10, 11, 13] with applications to working-memory load analysis.

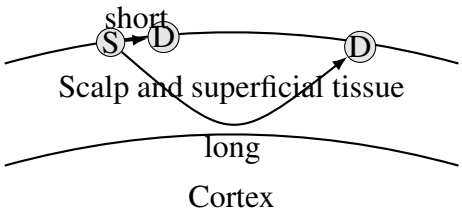


Fig. 4. Conceptual short- and long-separation fNIRS channels. Short channels estimate superficial physiology; long channels contain both superficial and cortical components.

2.6. Walking-Task Classification in Older Adults

Ma et al. classified walking task states in older adults with the help of fNIRS [3]. The paper is important in this respect that it shows the possibility of assessing complex neurophysiological functions related to mental and motor abilities in this age category. The authors have not made any attempt to establish the relation of these patterns to some clinical disorders; the experiment itself has merely confirmed the viability of task-based fNIRS analysis.

Image recognition components of the pipeline described in the article should not be used to reshape the thesis data into a standard generic image format. To the contrary, the better option would be to maintain a channel-by-time haemodynamic representation and validate models subject-wise.

2.7. Cognitive-Effort Modeling Framework

A cognitive-effort modeling framework is interesting because it allows accounting for individual effort spent on completing a task in addition to just the task-related changes [8]. Such individual effort expenditure may differ even in case when participants achieve the same results. This approach can contribute to mental-health analysis, although it needs further development.

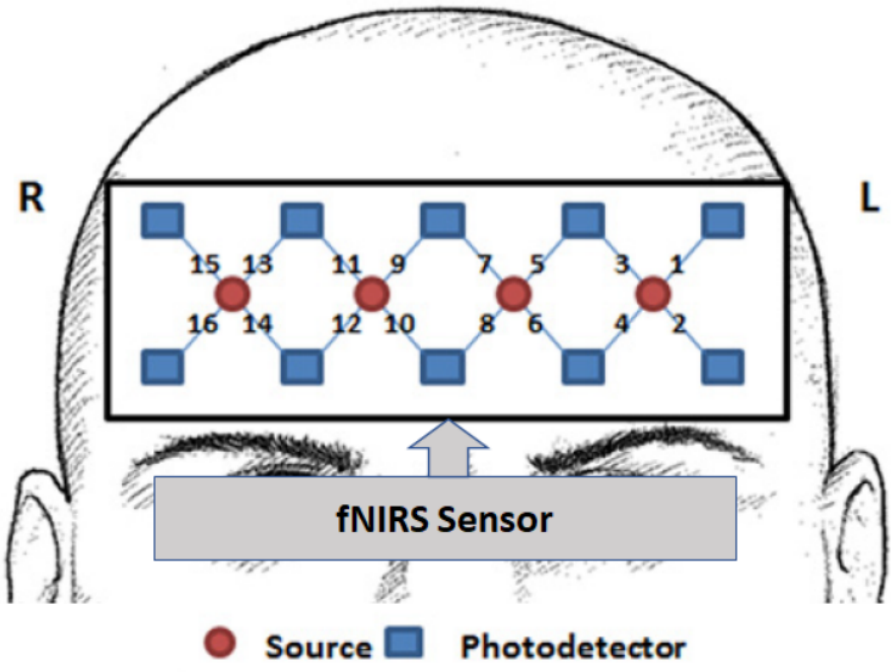


Fig. 5. Example fNIRS sensor placement used in walking-task classification [3].

Cognitive effort framework and evaluation metrics:

$$z = \frac{x-\mu}{\sigma} \quad \text{(Z-score standardization)} \tag{15}$$

$$P_z(s,i) = \frac{P(s,i)-\mu(P_i)}{\sigma(P_i)} \quad \text{(Participant-specific standardized performance)} \tag{16}$$

$$M_z(s,i) = \frac{M(s,i)-\mu(M_i)}{\sigma(M_i)} \quad \text{(Participant-specific standardized cognitive load)} \tag{17}$$

$$\text{RNE} = \frac{P_z - M_z}{\sqrt{2}} \quad \text{(Relative Neural Efficiency)} \tag{18}$$

$$\text{RNI} = \frac{P_z + M_z}{\sqrt{2}} \quad \text{(Relative Neural Involvement)} \tag{19}$$

Note: The concept of cognitive effort as such remains interesting; however, only relative haemodynamic data will be used from this thesis database.

3. Cross-Study Interpretation

In conclusion, the reviewed evidence suggests that a modest claim about fNIRS and machine learning can be made: the techniques may be used to classify task states and assess workload-related patterns, although clinically significant conclusions cannot yet be drawn [7, 4, 6].

Methodology-wise, short-channel regression becomes crucial, and a classifier input must be able to account for the temporal structure of the physiological signal. Finally, evaluation should prevent leakage from the same participant or overlapping windows.

3.1. Leakage-Safe Evaluation Pipeline

Physiological machine-learning classifiers are highly vulnerable to overfitting in case when windows from the same participant are contained in training and testing sets [15]. Instead of being trained on task-related patterns, such classifier is likely to learn stable participant-specific features or even motion artifacts. A leakage-free protocol for fNIRS classification studies was recently introduced as BenchNIRS [14].

Thus, in this thesis, any data splitting must be performed participant-wise before the training of the classifier. All hyperparameters should be tuned on training data only and validated on separate data. Scaling, feature selection, dimensionality reduction, and model selection steps must be done in the same manner. For data with windows, all windows from the same participant are supposed to stay in the same part of the split.

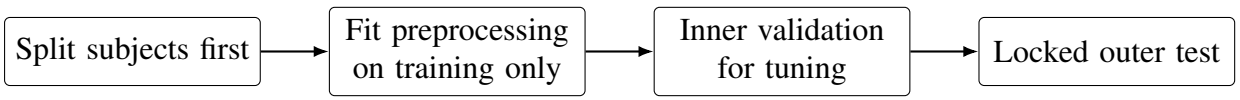


Fig. 6. Leakage-safe evaluation pipeline for fNIRS machine learning.

3.2. Methodological Implications

As for traditional machine-learning models, those are important as baselines. At least four of them should be tested: logistic regression or linear discriminant analysis, linear support vector machine, RBF-kernel SVM, and random forest. The CNNs, LSTM, CNN-LSTM, and temporal convolutional models can become useful only if they generalize better than the above models [14, 2, 9].

Furthermore, a proper model interpretability should be implemented. In case the predicted variable refers to mental or neurological condition, the thesis should discuss the contribution of particular channels, windows, or haemodynamic quantities.

4. Conclusions

This literature analysis shows that the thesis methodology can be supported by a modest, but solid claim: the combination of fNIRS and machine learning is currently more effective for classifying haemodynamic patterns than clinical diagnosis.

To that end, the updated methodology considers the following factors: short-distance channel regression for eliminating extracerebral effects, preservation of channel-by-time representation, distinction of relative haemodynamic patterns, and reporting of oxygenation only when explicitly estimated by the instrument and algorithm.

The future research should address leakage problems, employ nested hyperparameter optimization, perform preprocessing in a leakage-safe manner, use robust baselines, and implement model interpretability techniques.

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